

Teaching Exam

Subject – Computer Science

Topic – Computer Network



Lecture No.- 02

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Topics to be Covered

Topic

1:- Network Device ✓

2:- Types of Network ✓ and its application

3:- Advantage and ✓ Disadvantage of Network



Topic- Network Devices

Common Types of Networking Devices and Their Uses

Network devices work as a mediator between two devices for transmission of data, and thus play a very important role in the functioning of a computer network. Below are some common network devices used in modern networks:

- Access Point ✓
- Modems ✓
- Firewalls ✓
- Repeater ✓
- Hub ✓
- Bridge ✓
- Switch ✓
- Routers ✓
- Gateway ✓
- Brouter ✓
- NIC ✓

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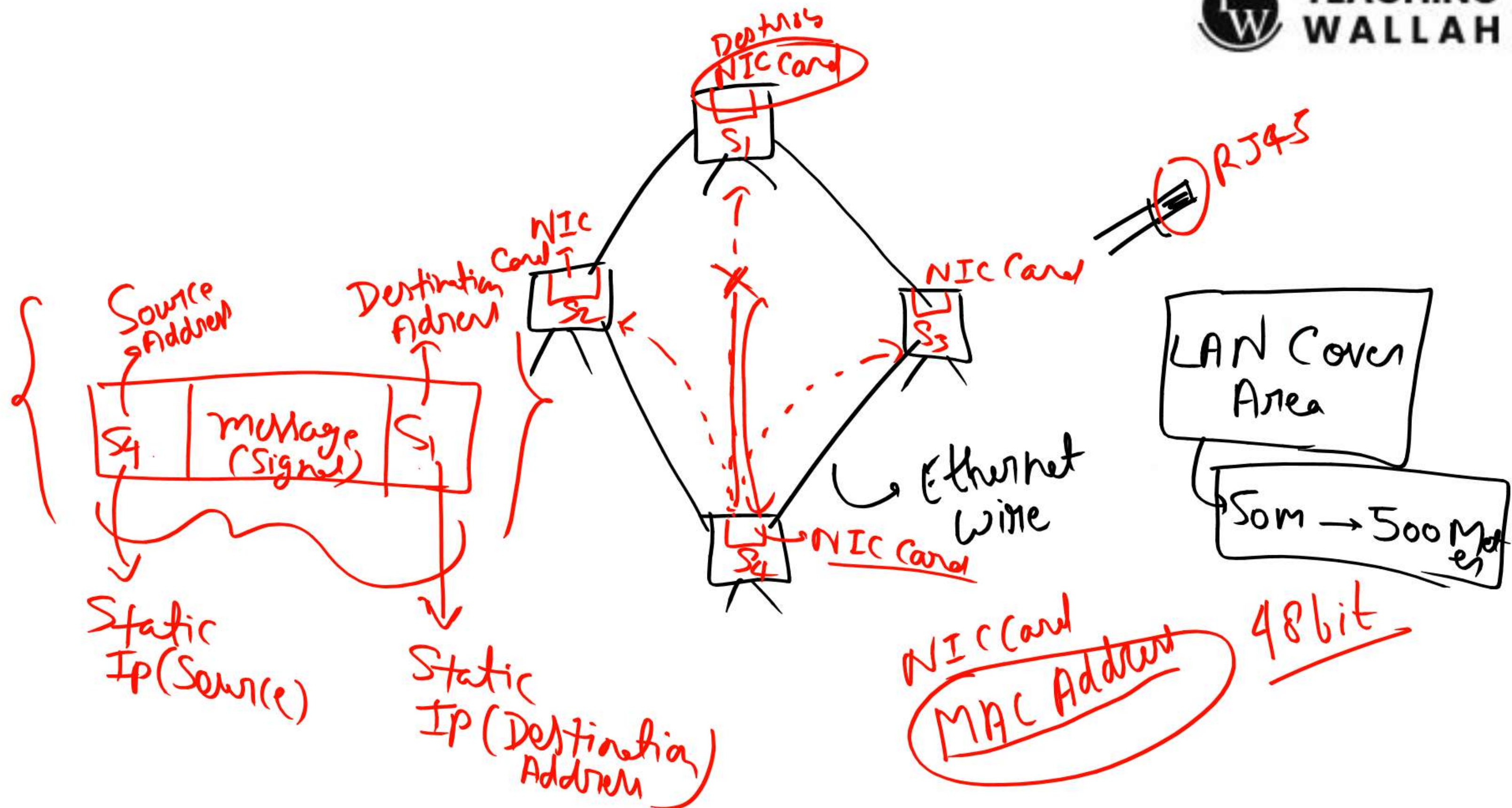
- Network Hardware
- **NIC CARD:-** It Stands for Network Interface Card (Hardware) attached to a host so as to establish network Connections.
- Every NIC Card has a unique physical address called MAC → address(Media Access Control Address) which is a 6-byte(48 bits) addressed assigned by the NIC Manufacturer.

Device Address

It is also called Network Interface Unit (NIU) or Terminal Point(TAP)

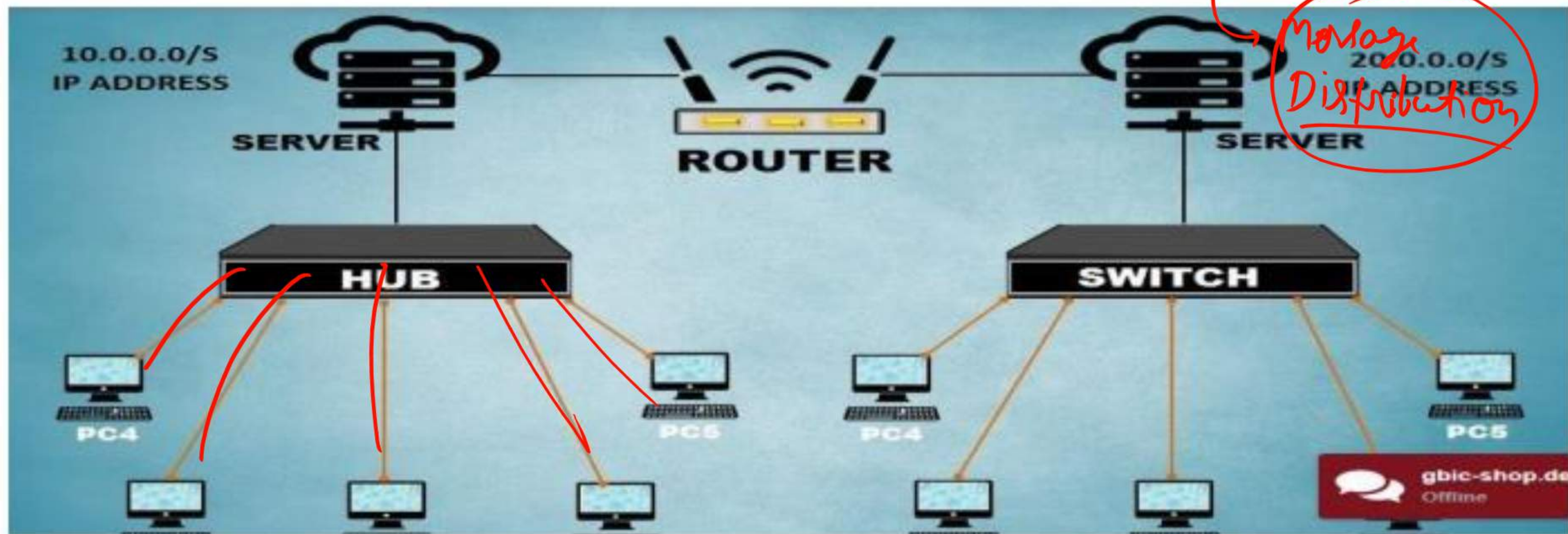
LAN → Local Area Network

NIC or network interface card is a network adapter that is used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique ID that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and the router or modem. NIC is a layer 2 device which means that it works on both the physical and data link layers of the network model.



Topic- Network Device - HUB

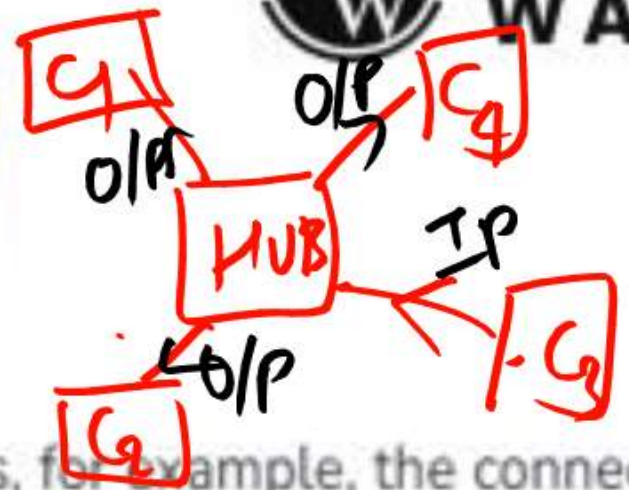
HUB & SWITCH



Topic- Network Device - HUB

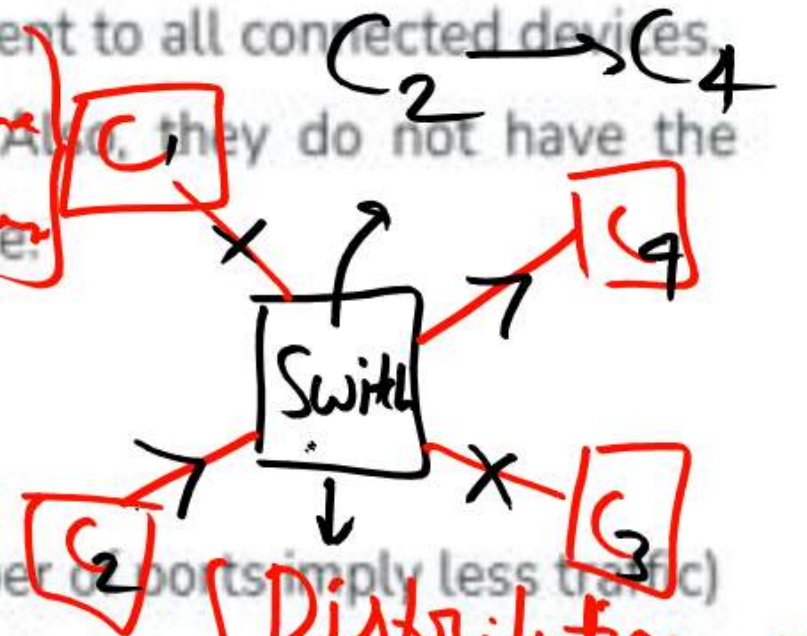
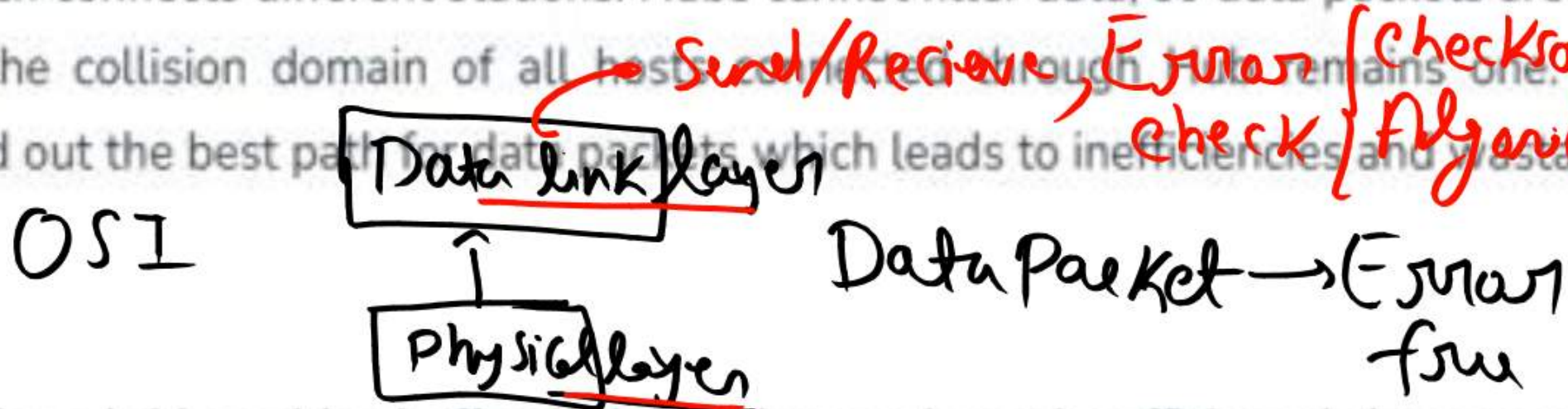
Hub

A hub is a multiport repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.



Switch

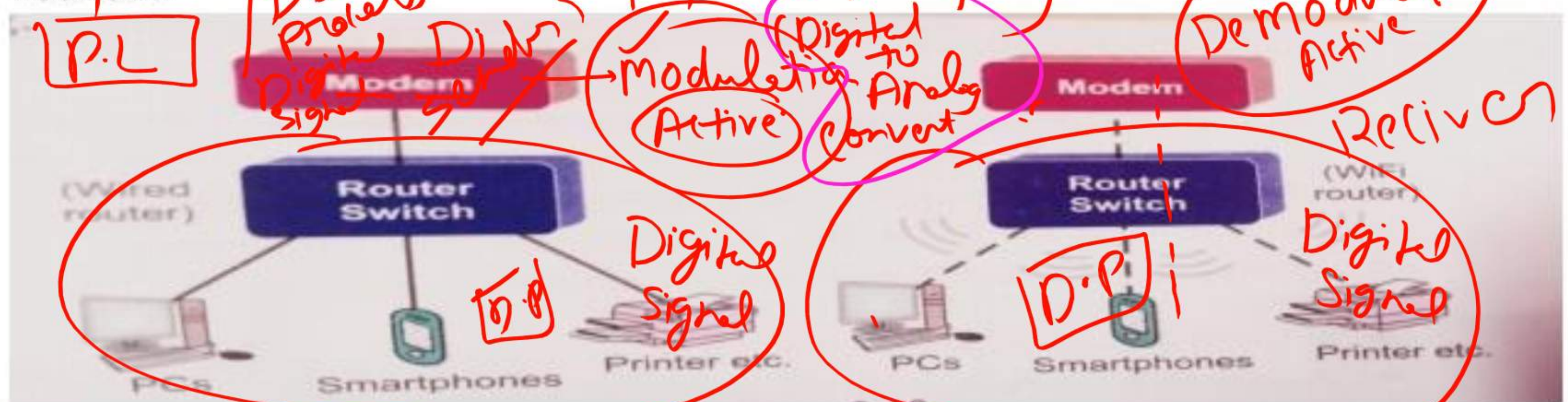
A switch is a multiport bridge with a buffer designed that can boost its efficiency (a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only. In other words, the switch divides the collision domain of hosts, but the broadcast domain remains the same.



Distribution Device
But Particular Device

Topic- Network Device - ROUTER

Router - The Router is responsible for Forwarding data from one Network to different network.



Router

A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.

Topic- Network Device – Access Point, Modems

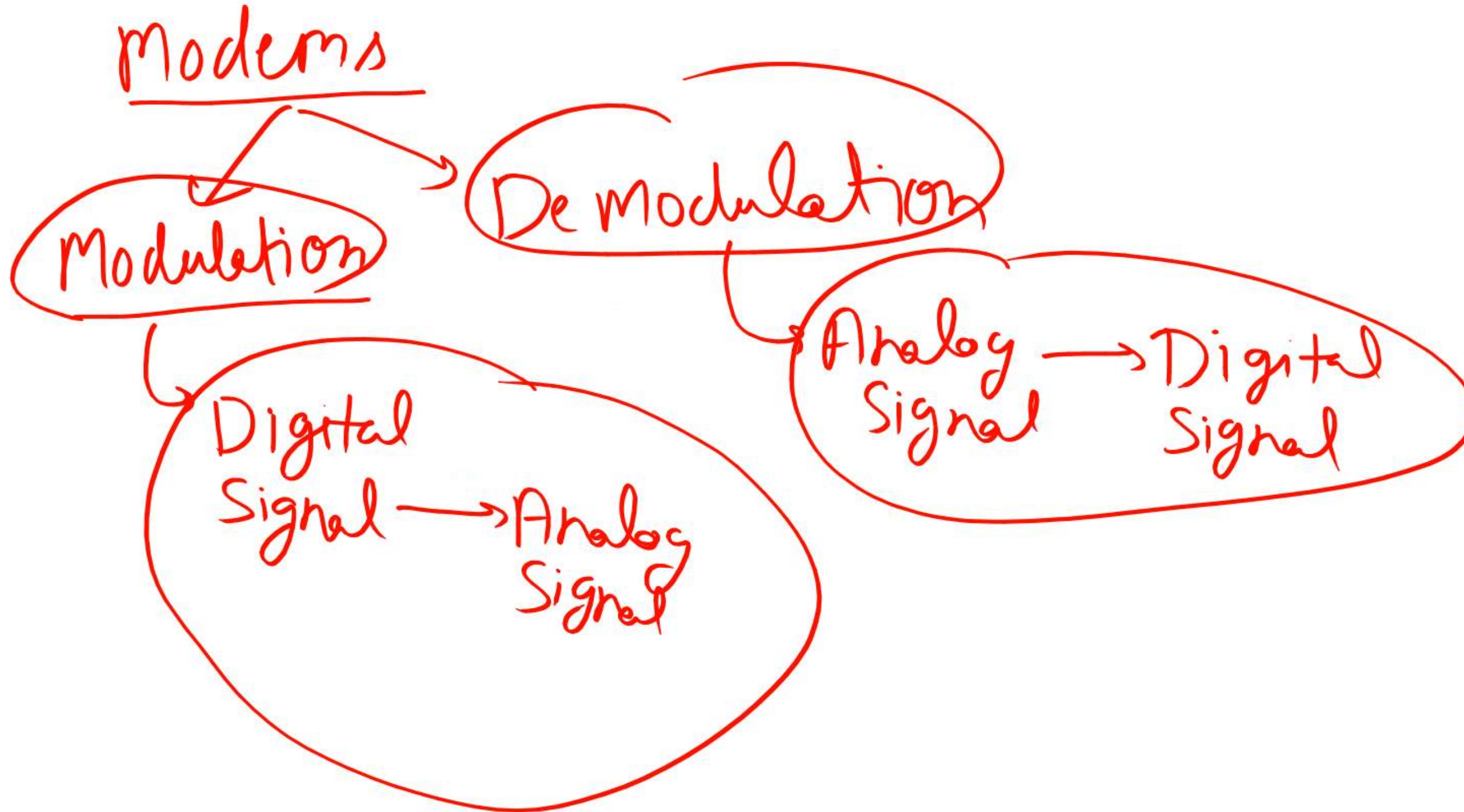
Access Point

An access point in networking is a device that allows wireless devices, like smartphones and laptops, to connect to a wired network. It creates a Wi-Fi network that lets wireless devices communicate with the internet or other devices on the network. Access points are used to extend the range of a network or provide Wi-Fi in areas that do not have it. They are commonly found in homes, offices, and public places to provide wireless internet access.

Modems

Modem is also known as modulator/demodulator is a network device that is used to convert digital signal into analog signals of different frequencies and transmits these signals to a modem at the receiving location. These converted signals can be transmitted over the cable systems, telephone lines, and other communication mediums. A modem is also used to convert an analog signal back into digital signal. Modems are generally used to access the internet by customers of an Internet Service Provider (ISP).





Topic- Network Device – Firewall , Repeater

Firewalls

A firewall is a network security device that monitors and controls the flow of data between your computer or network and the internet. It acts as a barrier, blocking unauthorized access while allowing trusted data to pass through. Firewalls help protect your network from hackers, viruses, and other online threats by filtering traffic based on security rules. Firewalls can be physical devices (hardware), programs (software), or even cloud-based services, which can be offered as SaaS, through public clouds, or private virtual clouds.

Repeater

A repeater operates at the physical layer. Its main function is to amplify (i.e., regenerate) the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network. When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting following the original strength. It is a 2-port device.



Security Bridge [Network
Save from Virus
& Threats]
Unauthorized blocked
Access

Topic- Network Device – GATEWAY

What is Gateway?

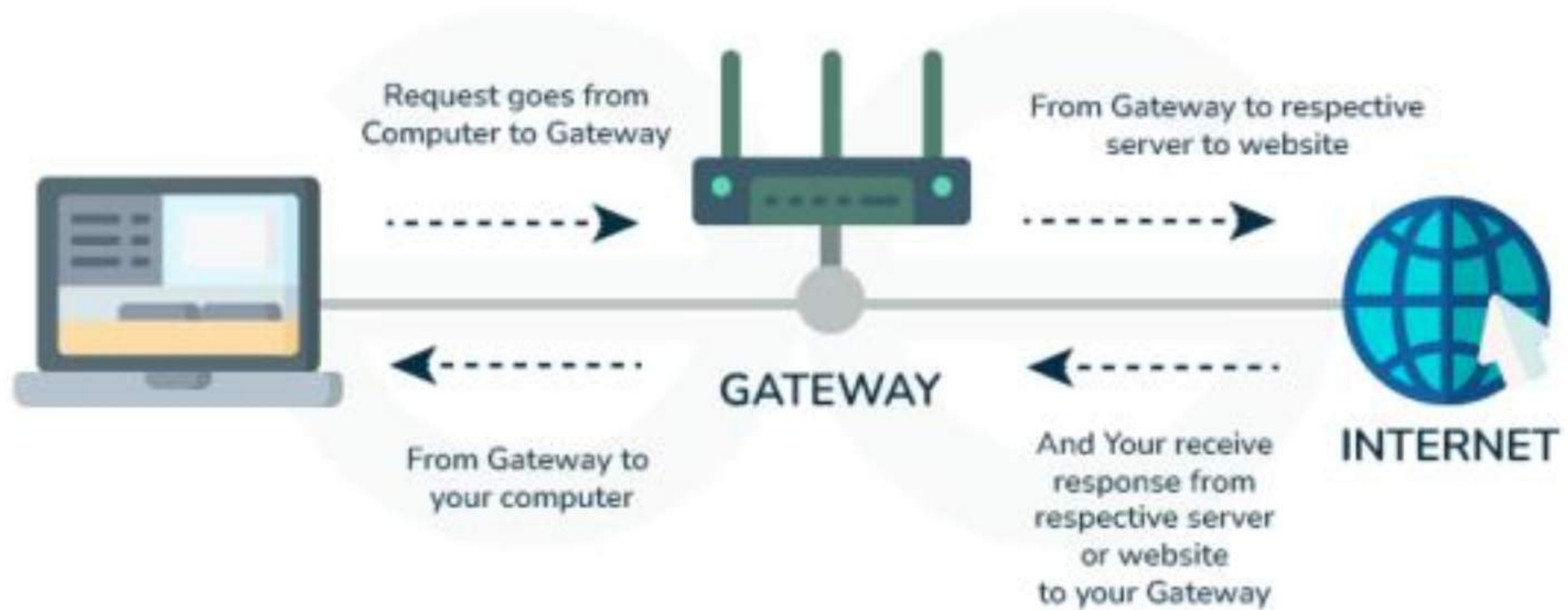
A gateway is a connecting point of any network that helps it to connect with different networks. The gateway monitors and controls all the incoming and outgoing traffic of the network. Suppose there are two different networks and they want to communicate with each other, so they need to set up a path between them. Now that path will be made between gateways of those different networks. Gateways are also known as protocol converters because they help to convert protocol supported by traffic of the different networks into that are supported by this network. Because of that, it makes smooth communication between two different networks.

How does Gateway Work?

Gateway has a simple working methodology of five steps:

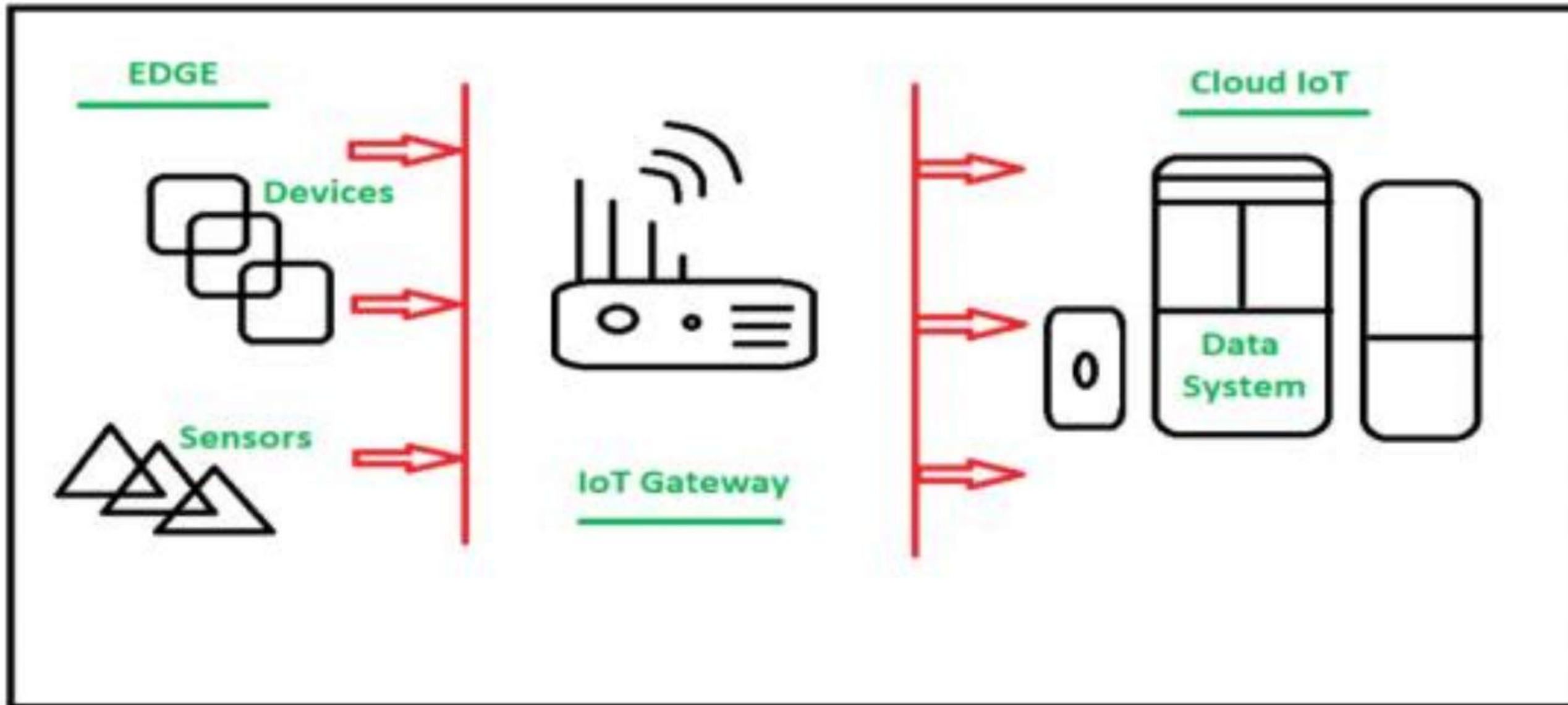
- **Step 1:** It gets data from the network
- **Step 2:** It intercepts and analyzes the received data.
- **Step 3:** It routes the data to the destination address.
- **Step 4:** It converts the received data to make that compatible with the receiver network.
- **Step 5:** It sends the final data inside the network.

Topic- Network Device – GATEWAY



Topic- Network Device – GATEWAY

Gateway is a device that connects different networks. In simpler terms, it is like a bridge. Its function is to connect the device in one network to interact with a device in another network whereas



Topic- Network Device – Difference between GATEWAY & ROUTER

Difference between Gateways and Router

Gateways	Router
A gateway is a device that is used for communication between networks having different sets of protocols.	Route is a device that receives, analyzes, and forwards the data packets to other networks.
Gateway connects two different networks.	It routes the data packets via the same networks.
Gateway does not support dynamic routing.	Router supports the dynamic routing.
The main function of a gateway is protocol translation.	The main function of a router is routing the traffic from one network to another network.
It is hosted on dedicated applications, physical servers, or virtual applications.	It is hosted on only the dedicated applications.
The gateway is also called a gateway router, proxy server, and voice gateway.	The router is also called a wireless router and an Internet router.

Topic- Network Device – Difference between GATEWAY & MODEM



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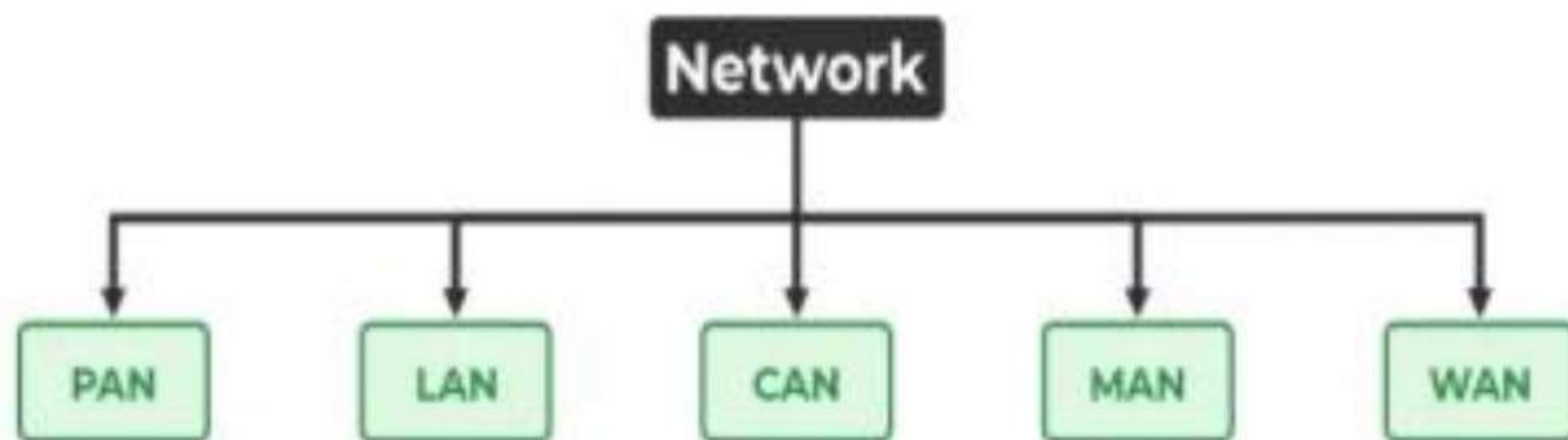
Basis of Comparison	Gateway	Modem
Definition	It is a device that is the center point for two or more networks to communicate with each other.	A device that connects a home or office device to the internet.
Function	Any device that enables traffic flow in and out of the network can function as a gateway.	It converts the digital signals to analog to transmit them in cable lines later converting them back to digital at the receiver.
Integration	It includes a router that directs traffic between devices on a network.	Does not include a router instead mainly on the connection of devices to the internet.
Protocols	May include protocols like a firewall, Wi-Fi access point, or Ethernet switch.	Supports one or two protocols such as cable or DSL.
Uses	Commonly used in businesses and large organizations.	Commonly used in homes.
Connection	A gateway connects to multiple devices.	A modem connects to a single device.
IP address management	A gateway can support multiple private IP addresses of multiple devices	A modem supports a single public IP address.
Cost	Gateway is expensive.	The modem is cheaper than Gateway.

Types of Computer Networks

Computer networks are classified based on several factors such as geographical area, ownership, architecture, topology, and transmission technology. The below is the classification of the networks :

Classification Based on Geographical Area

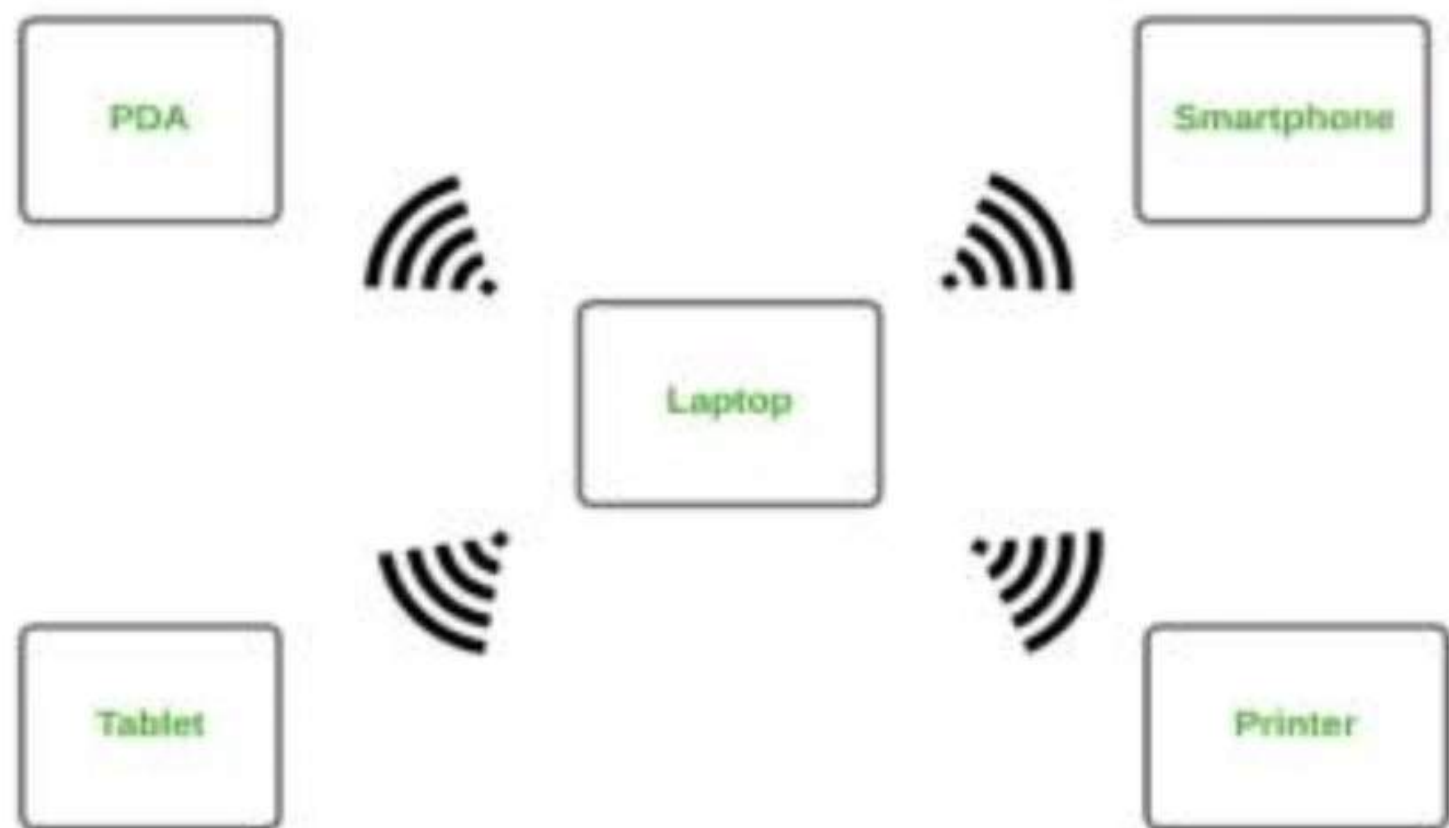
This is the most common way of classifying networks below are the different types :



1. Personal Area Network (PAN)

PAN is the most basic type of computer network. It is a type of network designed to connect devices within a short range, typically around one person. It allows your personal devices, like smartphones, tablets, laptops, and wearables, to communicate and share data with each other. PAN offers a network range of 1 to 10 meters from person to device providing communication. Its transmission speed is very high with very easy maintenance and very low cost.

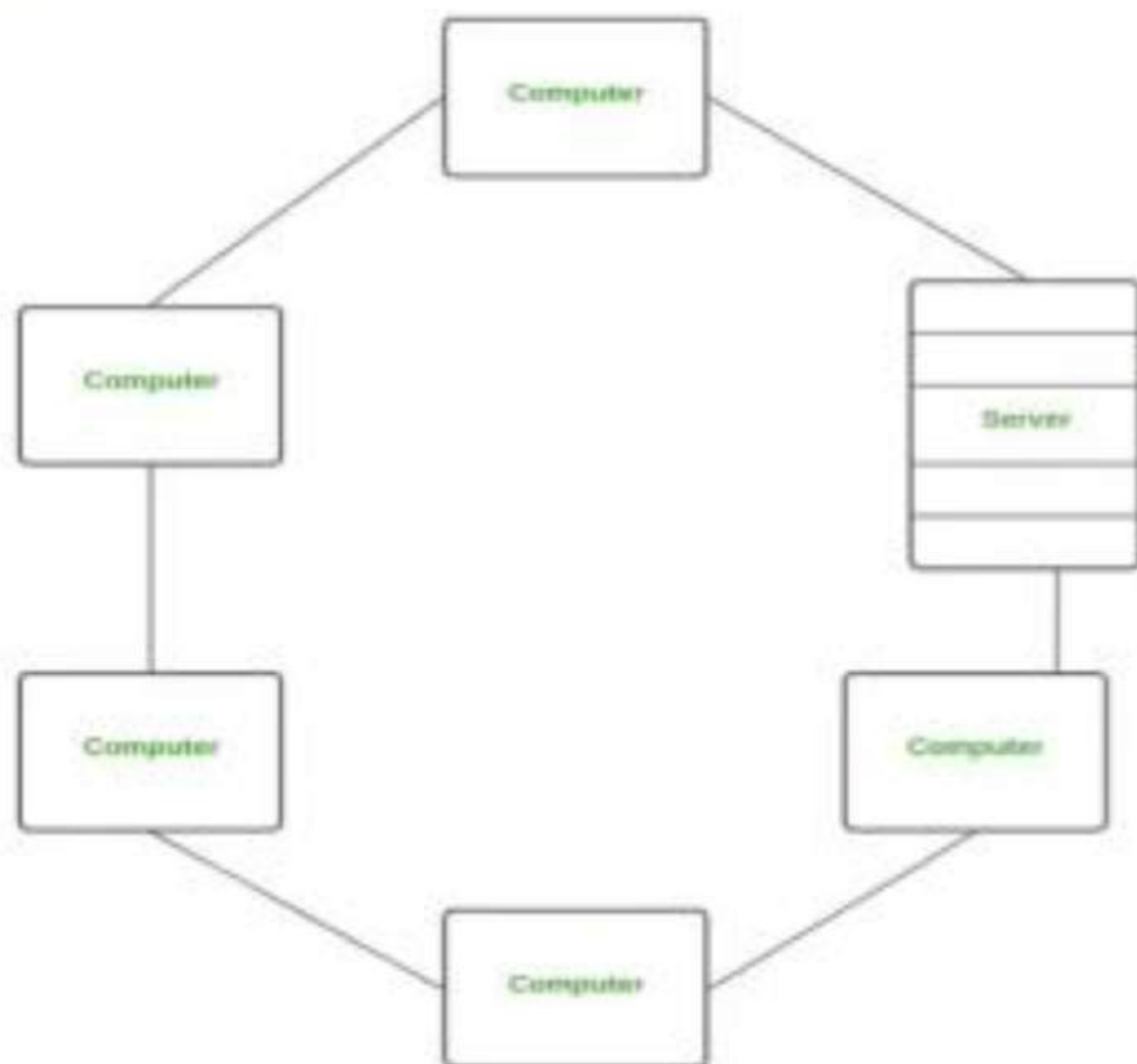
Examples of PAN are Bluetooth connection between a phone and wireless earbuds , Infrared communication between TV and remote.



2. Local Area Network (LAN)

LAN is the most frequently used network. It is a computer network that connects computers through a common communication path, contained within a limited area, that is, locally. A LAN encompasses two or more computers connected over a server. The two important technologies involved in this network are Ethernet and Wi-fi. It ranges up to 2km & transmission speed is very high with easy maintenance and low cost.

Examples of LAN are Wi-Fi in a home or school, wired LAN in a company's office.

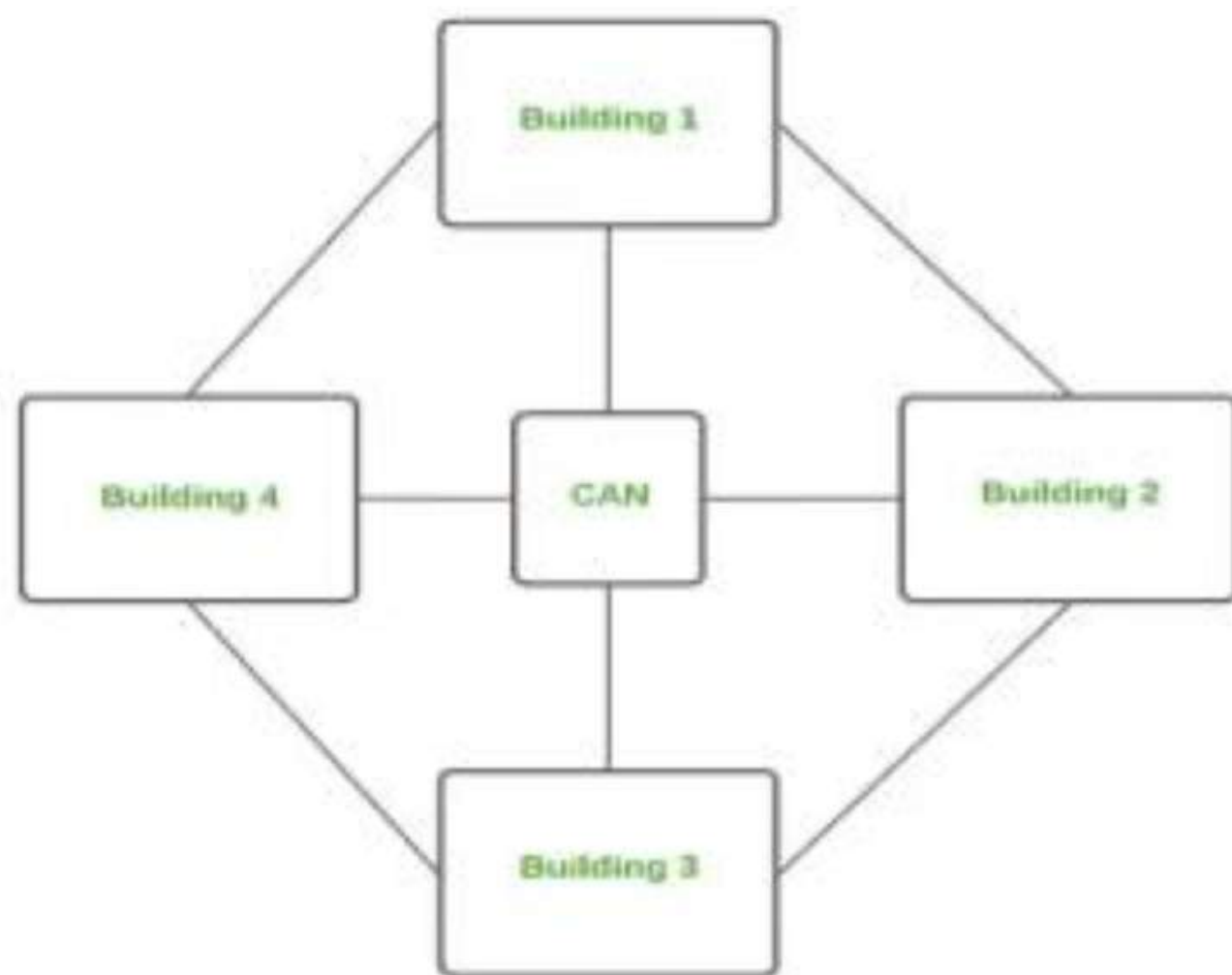


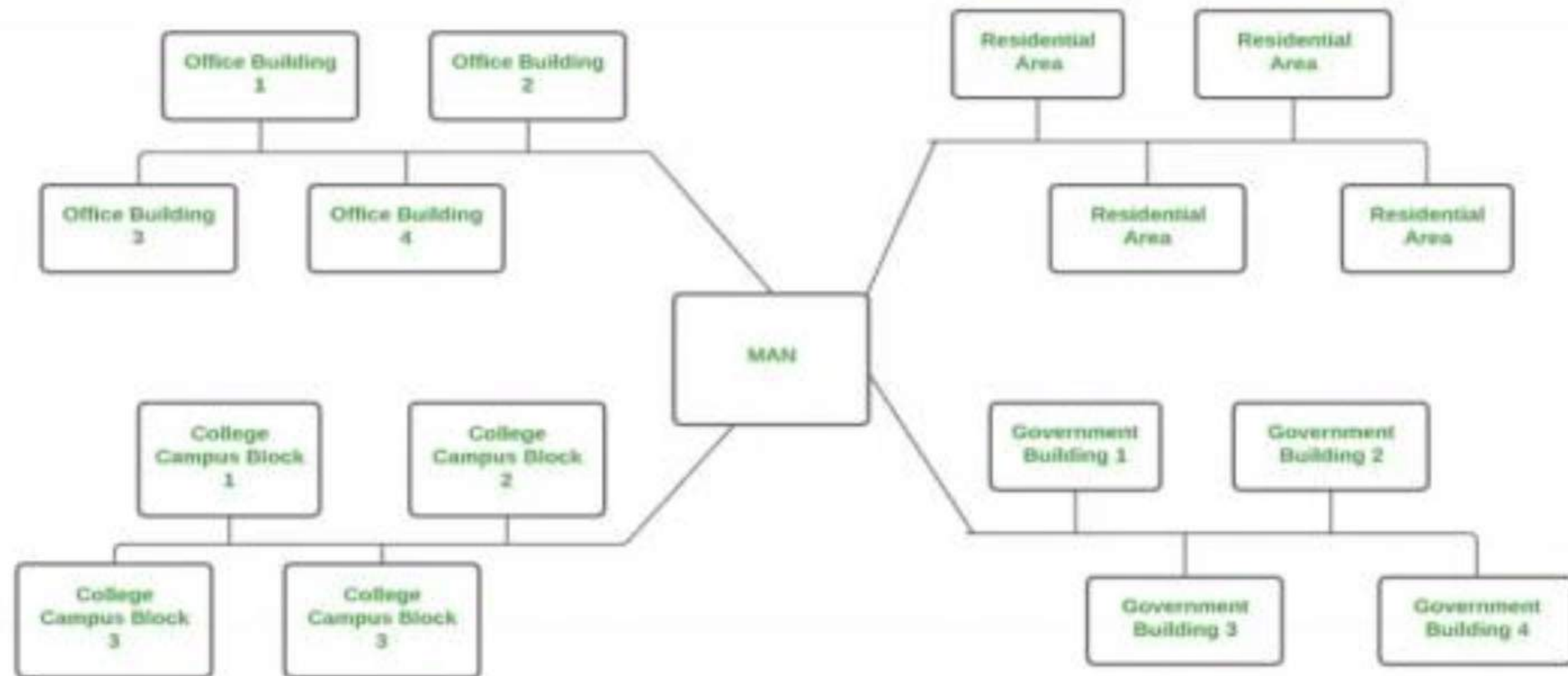
Local Area Network

3. Campus Area Network (CAN)

CAN is bigger than a LAN but smaller than a MAN. This is a type of computer network that is usually used in places like a school or colleges. This network covers a limited geographical area that is, it spreads across several buildings within the campus. CAN mainly use Ethernet technology with a range of few kilometers. Its transmission speed is very high with a moderate maintenance cost and moderate cost.

Examples of CAN are networks that cover schools, colleges, buildings, etc.

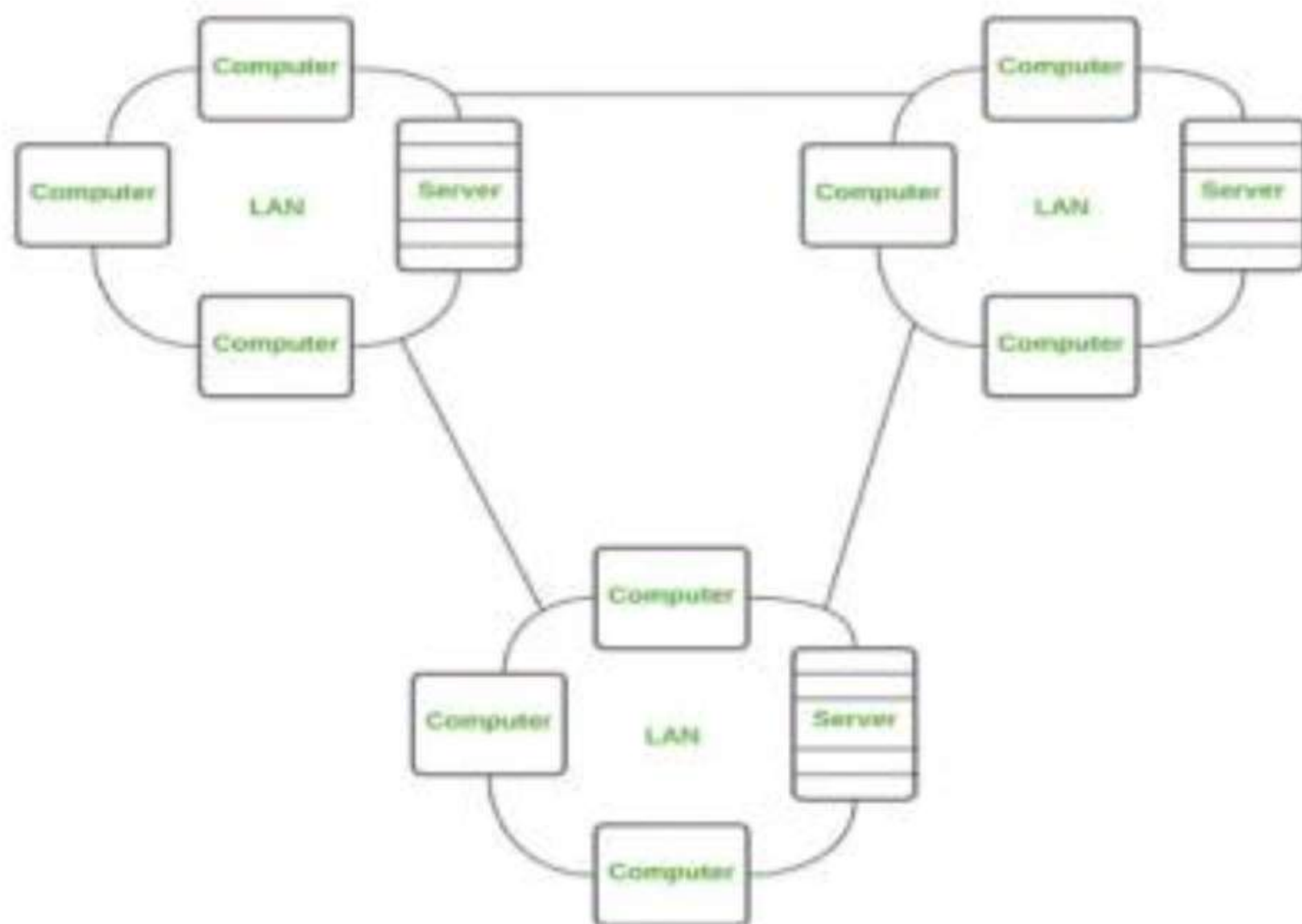




5. Wide Area Network (WAN)

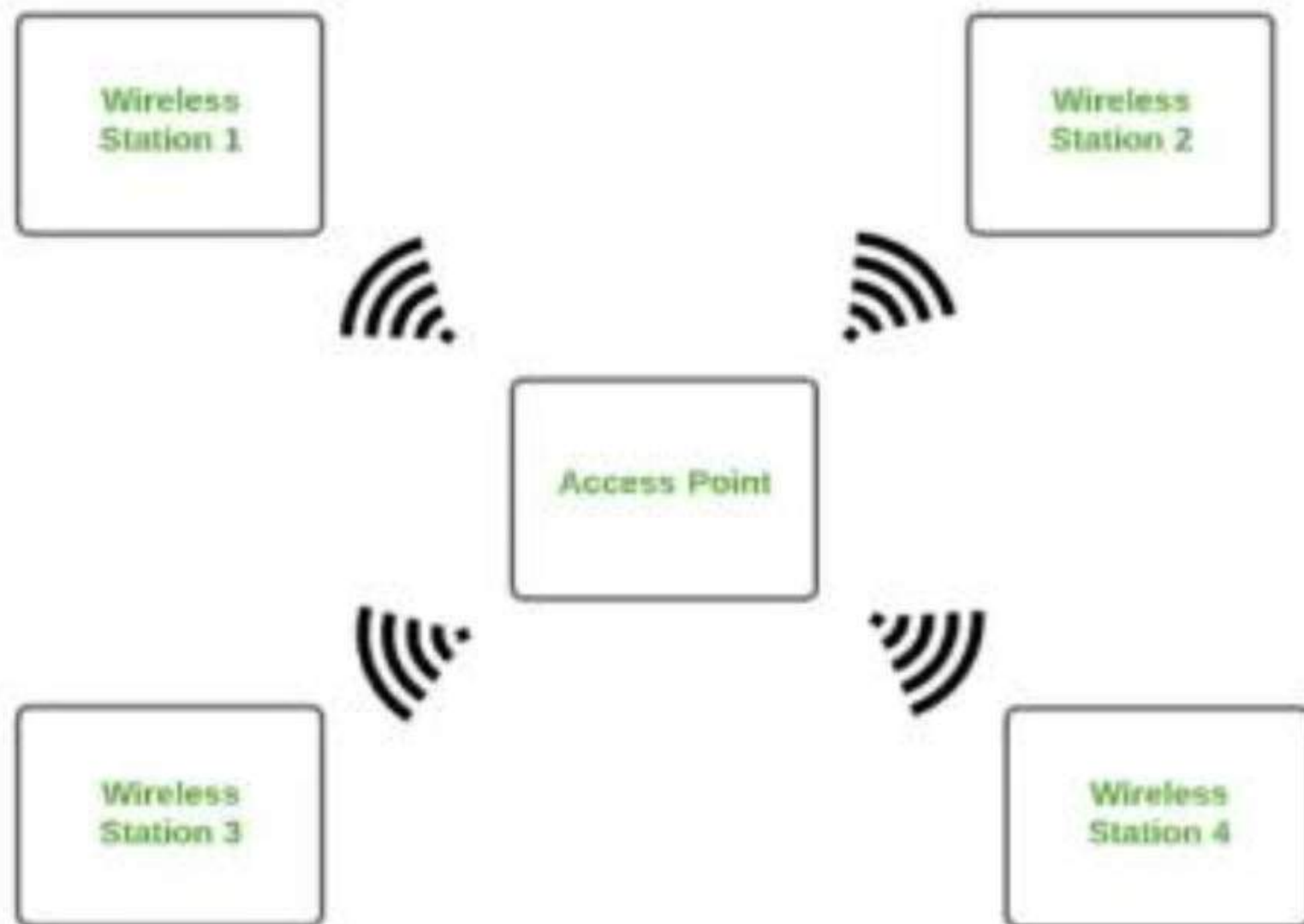
WAN is a type of computer network that connects computers over a large geographical distance through a shared communication path. It is not restrained to a single location but extends over many locations. WAN can also be defined as a group of local area networks that communicate with each other with a range above 50km. Here we use Leased-Line & Dial-up technology. Its transmission speed is very low and it comes with very high maintenance and very high cost.

Examples of WAN are the Internet (largest WAN), Banking networks linking global branches



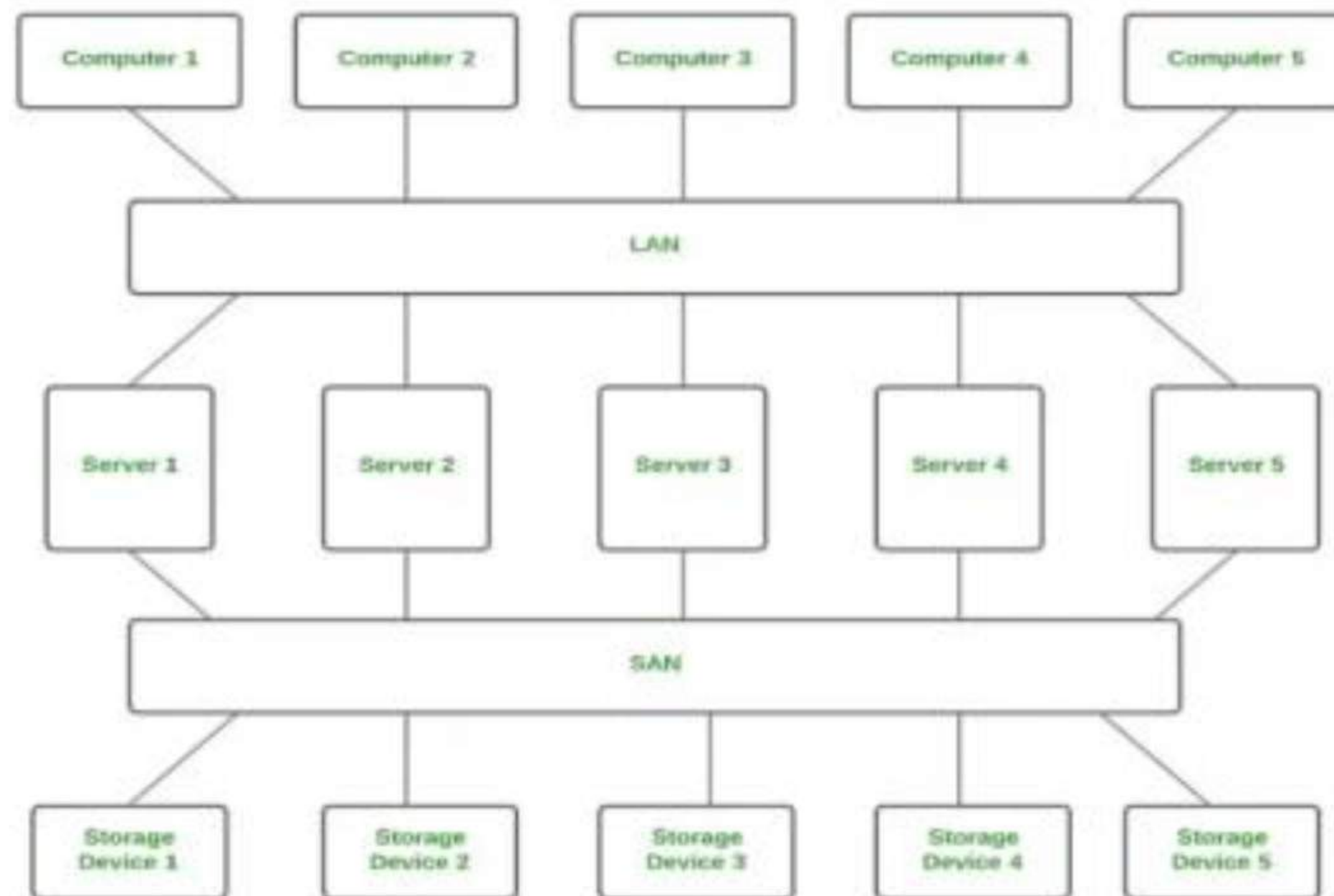
1. Wireless Local Area Network (WLAN) :

WLAN is a type of computer network that acts as a local area network but makes use of wireless network technology like Wi-Fi. This network doesn't allow devices to communicate over physical cables like in LAN but allows devices to communicate wirelessly. The most common example of WLAN is Wi-Fi.



2. System Area Network (SAN)

A System Area Network is designed to connect high-performance computers within a localized, high-speed environment, such as in data centers or supercomputing facilities. A SAN provides access to block-level data storage. Examples of SAN are a network of disks accessed by a network of servers.



Why storage area networks are important

- Computer memory and local storage resources might not provide enough storage, storage protection, multiple-user access or speed and performance for enterprise applications. So, most organizations employ some form of SAN in addition to network-attached storage (NAS) for improved efficiency and better data management.

Traditionally, only a limited number of storage devices could attach to a server, limiting a network's storage capacity. However, a SAN introduces networking flexibility that enables one server, or many heterogeneous servers across multiple data centers, to share a common storage utility. The SAN eliminates the traditional dedicated connection between a network file server and storage—and the concept that the server effectively owns and manages the storage devices—eliminating bandwidth bottlenecks. A SAN eliminates single points of failure to enhance storage reliability and availability.

A SAN is also optimal for disaster recovery (DR) because a network might include many storage devices, including disk, magnetic tape and optical storage. The storage utility might also be located far from the servers that it uses.

1. Private Network

These are networks completely owned and managed by a single organization or individual. The owner controls who can connect, what they can do, and how data moves within the network. Since there is no outside access, they are highly secure and reliable, often protected by firewalls and strict policies.

Examples:

- A company's internal office network (Intranet)
- School or college campus networks
- Hospital systems storing patient data

2. Public Network

These are networks open to the general public and are usually provided by Internet Service Providers (ISPs) or businesses in public places. They offer basic or no authentication, making them less secure and vulnerable to attacks if not used carefully. Examples: Public Wi-Fi at cafes or airports, free hotspots in city centers, Internet (the largest public network).

3. Hybrid Network

A hybrid network blends private and public access, offering flexibility and role-based access control. Some parts are restricted (like internal systems), while others are open (like guest Wi-Fi). This setup is useful in environments where different users need different access levels. Example: A university network with private access for staff and students and limited access for guests.

Internetwork

An internetwork is a setup where two or more networks (like LANs or WANs) are interconnected using routers or gateways, allowing communication between them. This process is called internetworking.

There are two main types:

1. Intranet

An intranet is a private network used within an organization. It allows employees to access internal tools, documents, and systems securely. It's not available to the public, and typically uses private IP addresses. **Examples:** Company HR portals, Internal knowledge bases, Staff-only communication tools.

2. Extranet

An extranet is an extension of the intranet that provides limited, secure access to outsiders such as partners, vendors, or clients. It's used to collaborate across organizations while maintaining internal security. **Examples:** Supplier portals, Client dashboards for project updates, Partner login systems for shared services

Advantages of Computer Network

- **Central Storage of Data:** Files are stored on a central storage database which helps to easily access and available to everyone.
- **Connectivity:** A single connection can be routed to connect multiple computing devices.
- **Sharing of Files:** Files and data can be easily shared among multiple devices which helps in easily communicating among the organization.
- **Security through Authorization:** Computer Networking provides additional security and protection of information in the system.

Disadvantages of Computer Network

- **Virus and Malware:** A virus is a program that can infect other programs by modifying them. Viruses and Malware can corrupt the whole network.
- **High Cost of Setup:** The initial setup of Computer Networking is expensive because it consists of a lot of wires and cables along with the device.
- **loss of Information:** In case of a System Failure, might lead to some loss of data.
- **Management of Network:** Management of a Network is somehow complex for a person, it requires training for its proper use.

THANK YOU